MARINDUQUE STATE UNIVERSITY

College of Information and Computing Sciences

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DELIVERABLE #7

G-TRAMS

GASAN TRICYCLE REGULATORY & AUTOMATED MANAGEMENT SYSTEM

USER'S MANUAL OF THE SYSTEM

Course / Program:	BS Information Technology (Capstone Project)
System Version:	v2.4.0 (Production Release)
Deployment URL:	https://g-trams-official.vercel.app
Development Team:	Group 5 (CICS)
Lead Backend:	Jhon Kyn Axix H. Cabrigas
Frontend Lead:	Justine S. Lachica
Quality Assurance:	Jay Vincent G. Motol
DevOps & Cloud:	Ian Simon Francis M. Rey

Municipality of Gasan, Province of Marinduque • Academic Year 2026–2027

1. SYSTEM OVERVIEW

The **Gasan Tricycle Regulatory and Automated Management System (G-TRAMS)** is an enterprise-grade, cloud-native web portal developed for the Local Government Unit (LGU) of Gasan, Marinduque. It replaces outdated, paper-intensive tricycle franchising procedures with a fully synchronized, role-based digital architecture.

1.1 Purpose and Objectives

- **Automate MTOP Issuance:** Streamline the entire lifecycle of Motorized Tricycle Operator's Permits (MTOP) from online application, verification, fee settlement, to digital permit issuance.
- **Eliminate "Colorum" Units:** Enforce strict municipal body number assignments, preventing vehicle duplicates, forged documents, and unregistered commercial operations.
- **Integrate Roadside Enforcement:** Equip municipal police and traffic enforcers with mobile QR code scanning tools to verify legitimate permits in under 3 seconds.
- **Promote Financial Transparency:** Provide the Municipal Treasury with real-time payment audit trails, instant official receipting, and COA-compliant daily remittance sheets.
- **Comply with National Standards:** Ensure full adherence to the *Data Privacy Act of 2012 (RA 10173)*, ARTA Ease of Doing Business regulations, and ISO-standard local government recordkeeping.

1.2 Intended User Roles

User Role	Description & Scope of Authority	Primary Workflows
Tricycle Operator / Driver	Individual tricycle franchise holders and drivers residing in Gasan.	Online franchise filing (New/Renewal/Dropping), document upload, application tracking, digital ID/certificate viewing.
TODA President	Elected leaders of accredited Tricycle Operators and Drivers Associations.	Master roster management, individual member addition, digital clearance issuance, and member removal requests.
Traffic Enforcer (PNP / CTMO)	Field traffic officers authorized to inspect tricycles on municipal routes.	Mobile QR code verification of operator IDs and vehicle MTOPs, issuing digital traffic citation tickets.
Municipal Cashier / Treasury	Municipal Treasury disbursing officers handling revenue collection.	Payment assessment, logging cash and digital payments, issuing official receipts, and generating daily remittance sheets.
BPLO Licensing Officer	Business Permits and Licensing Office regulatory staff.	Queue review, document inspection with AI OCR extraction, physical inspection scheduling, and permit approval.

User Role	Description & Scope of Authority	Primary Workflows
System Administrator (IT)	Municipal IT officers responsible for system availability and security.	User access control (RBAC), audit logging, and automated weekly database backups with disaster recovery.



Figure 1. G-TRAMS Public Citizen Portal & Services Landing Page

2. SYSTEM REQUIREMENTS AND ACCESS

G-TRAMS operates as a lightweight Progressive Web Application (PWA) accessible via any modern web browser without requiring dedicated software installations.

2.1 Hardware and Software Requirements

Component	Minimum Requirement	Recommended Specification
Devices Supported	Android / iOS Smartphone, Tablet	Desktop PC or Laptop (1080p Display) for LGU Staff
Web Browser	Chrome 100+, Edge 100+, Safari 15+, Firefox 100+	Latest Google Chrome or Microsoft Edge
Internet Connection	Stable 3G/4G Mobile Data (minimum 1 Mbps)	Broadband / Wi-Fi (5 Mbps or higher)
Official URL	https://g-trams-official.vercel.app	

3. GETTING STARTED: ACCESS, REGISTRATION & LOGIN

3.1 Account Registration Procedure

Purpose: Allows new tricycle operators and TODA leaders to create a verified personal account.

1. Open your browser and navigate to **https://g-trams-official.vercel.app**.
2. Click the **"Register"** link located on the top navigation bar or below the login form.
3. Complete the registration form with your **Full Legal Name**, **Primary Mobile Number**, **Email Address**, and **Barangay Address** in Gasan.
4. Provide a secondary **Recovery Contact** (backup phone number or email) to ensure account recovery if your primary SIM or email is misplaced.
5. Select your designated **TODA Association** (or "NON-TODA" if unaffiliated) and vehicle body number.
6. Create a secure password containing at least 8 characters, an uppercase letter, and a number.
7. Click **"Create Account"** and enter the 6-digit OTP code received on your registered contact.

✓ Expected Result: Account is successfully registered, verified, and redirected to your role-specific dashboard.

Figure 2. G-TRAMS New User Account Registration with Recovery Contacts

3.2 Portal Login Procedure

Purpose: Grants authenticated access to authorized modules based on role permissions.

1. Navigate to **https://g-trams-official.vercel.app/login**.
2. Enter your registered **Email Address** or **Mobile Phone Number**.
3. Type your password in the password field. You may click the eye icon to verify visibility.
4. Click the **"Log In"** button.

✓ Expected Result: The system authenticates your token and directs you immediately to your active dashboard.

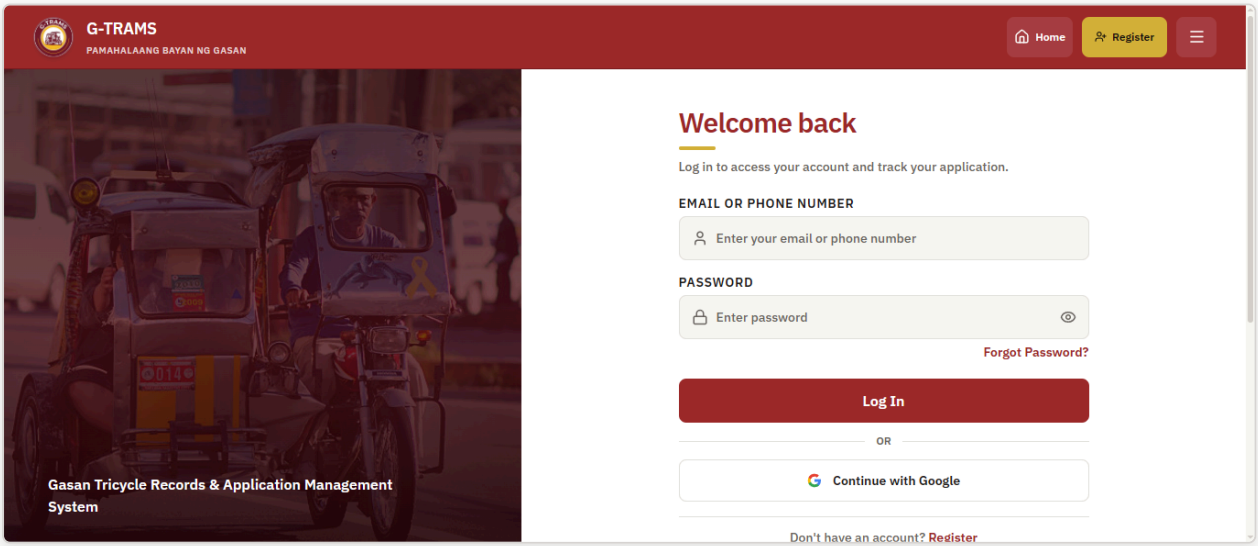


Figure 3. Official G-TRAMS Secure Authentication Login Terminal

3.3 Password Recovery (Forgot Password)

Purpose: Enables operators to regain account access via SMS or email verification.

1. On the login page, click **"Forgot Password?"**.
2. Enter your registered primary email, phone number, or backup recovery contact.
3. Click **"Send Reset Code"** to receive an authorized 6-digit one-time PIN.
4. Enter the received OTP code and configure your new secure password.

✓ Expected Result: Password is updated securely, allowing immediate sign-in with your new credentials.

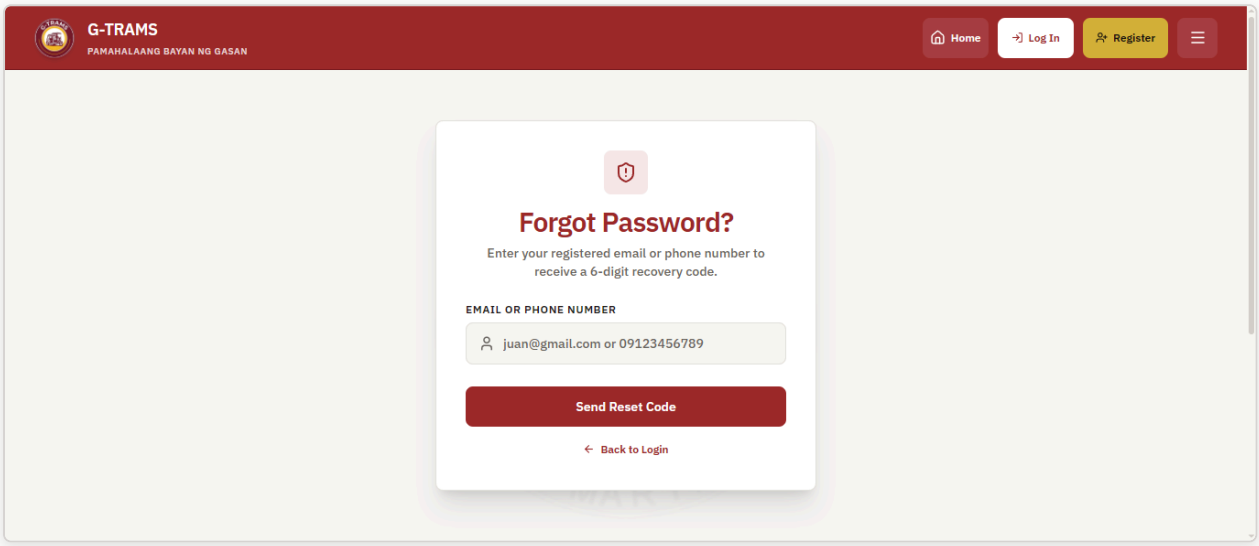


Figure 4. Password Recovery & Multi-Channel OTP Reset Interface

4. OPERATOR PORTAL: STEP-BY-STEP INSTRUCTIONS

The Operator Portal is designed for ease of use by tricycle owners and drivers. It presents a clear overview of active franchises, inspection dates, and permit credentials.

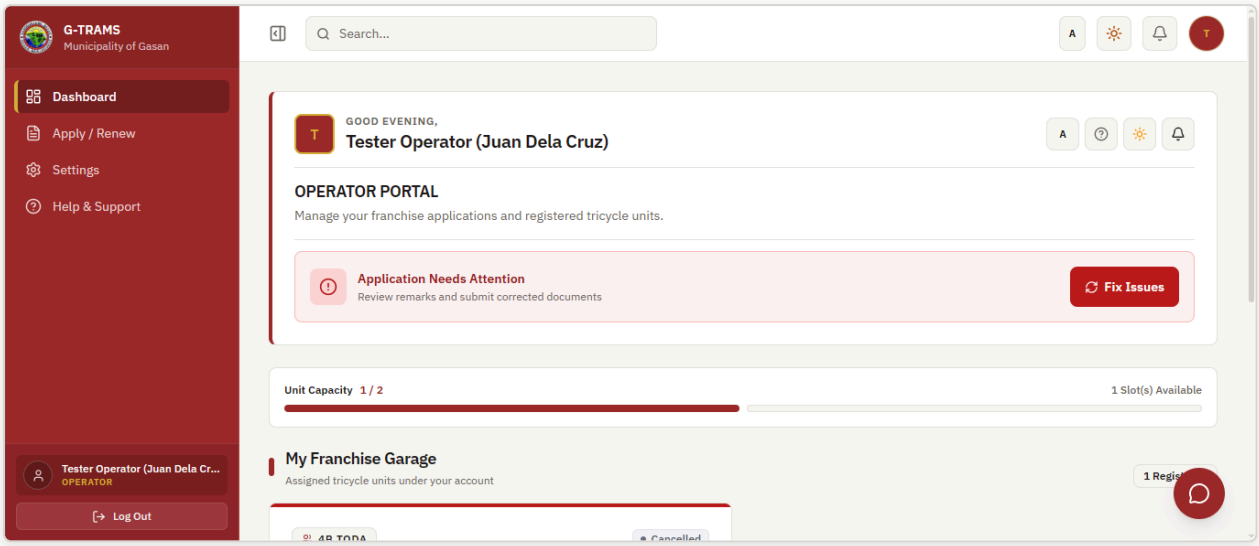


Figure 5. Operator Dashboard with Unit Allocation & Digital ID Badge

4.1 Submitting an Online Franchise Application

Purpose: Apply for a New Franchise, Annual Renewal, Dropping, or Transfer without visiting the municipal hall.

1. From your Operator Dashboard, click the **"Apply for Franchise"** or **"Apply Online"** button.
2. Select the application type: **New Franchise**, **Renewal**, or **Transfer of Ownership**.
3. Input the tricycle vehicle technical details:
   - **LTO Plate Number** (e.g., 123-ABC)
   - **Motor Number** & **Chassis Number** (from vehicle registration)
   - **Make / Brand** (e.g., Honda TMX 125, Kawasaki Barako)
   - **Assigned Body Number**
4. Upload clear digital photos or scans of mandatory credentials:
   - **LTO Official Receipt & Certificate of Registration (OR/CR)**
   - **Community Tax Certificate (Cedula)** with serial number and date
   - **Valid Professional Driver's License**
   - **TODA Clearance & Certification** issued by your TODA President
5. Click **"Submit Application"**. The system's integrated AI OCR engine will immediately inspect document text clarity and queue the submission for BPLO verification.

✓ Expected Result: Application status transitions to "*Pending Review*" with a generated application reference ID.

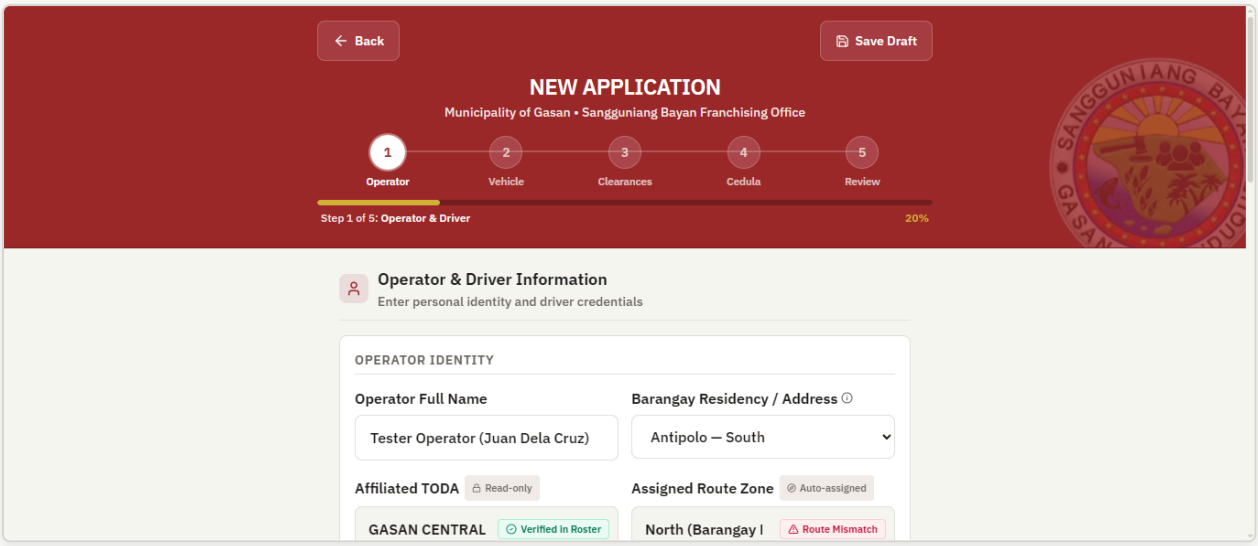


Figure 6. Online MTOP Application Form with Document Upload & OCR Pre-Check

4.2 Accessing Digital Operator ID & MTOP Certificate

Purpose: Display official digital credentials on mobile devices for inspection by authorities.

1. Log in to your Operator account and locate your approved unit under **"Active Franchises"**.
2. Click **"View Digital ID"** to open the full-screen official digital identification card.
3. To view or print your MTOP permit certificate, click **"Download / View MTOP Certificate"**.
4. Present the tamper-evident QR code directly to any municipal traffic officer during roadside inspection.

✓ Expected Result: Full digital permit displays with certified municipal seals, validity date, and route authorization.

5. TODA PRESIDENT PORTAL: ROSTER & CLEARANCES

TODA Presidents possess delegated authority to endorse members, manage association roster allocations, and ensure that only accredited operators receive barangay franchise privileges.

5.1 Managing Association Member Rosters

Purpose: Register new association members and verify compliance before LGU submission.

1. Log in with your verified TODA President credentials.
2. Navigate to the **"Submit / Manage Members"** module from the sidebar.
3. To register a single new member: Click **"Add Member"**, input the member's full name, license number, and assign a unique association body number. The system automatically blocks duplicate numbers.
4. To upload multiple members: Click **"Bulk Roster Upload"** and select your association's masterlist spreadsheet or photo.
5. Review each member's status and click **"Endorse / Issue Clearance"** to authorize their municipal MTOP application.

✓ Expected Result: Endorsed members immediately appear in the BPLO verification queue with validated TODA accreditation.

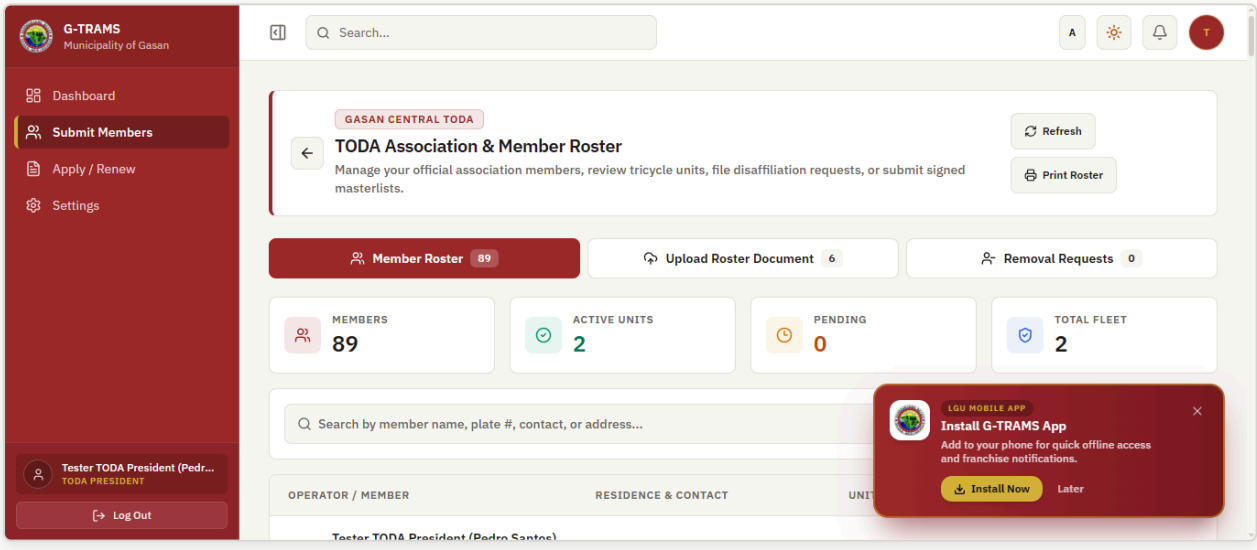


Figure 7. TODA President Member Roster Management & Body Number Assignment

6. TRAFFIC ENFORCER PORTAL: ROADSIDE INSPECTION

Field traffic enforcers and PNP personnel utilize mobile devices to scan tricycle QR codes and verify franchise validity instantaneously.

6.1 Scanning QR Codes and Roadside Verification

Purpose: Instantly detect illegal colorum vehicles, expired permits, and unauthorized drivers.

1. Open any mobile camera or QR scanner and point it at the physical tricycle MTOP sticker or driver's digital ID.
2. The browser opens the secure verification URL: `https://g-trams-official.vercel.app/verify/:id`.
3. Inspect the verification display:
   - **Franchise Status Badge:** Green (Active / Valid), Yellow (Renewal Pending), or Red (Expired / Revoked).
   - **Verified Vehicle Specs:** Plate Number, Motor Number, Body Number, and Registered TODA.
   - **Operator Identity:** Full legal name and registered contact number.

✓ Expected Result: Official LGU verification is displayed within 1 to 2 seconds, confirming genuine permit records.

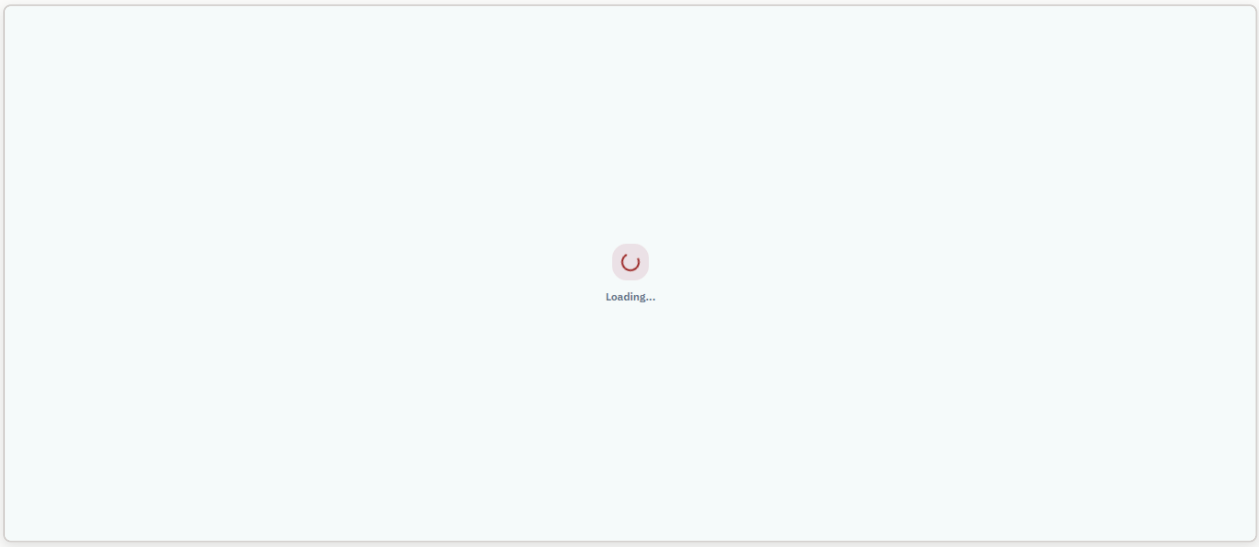


Figure 8. Live Real-Time Roadside QR Code Verification Terminal

7. MUNICIPAL CASHIER & TREASURY PORTAL

The Cashier Portal is an isolated financial terminal where treasury personnel assess regulatory fees, log official receipts (OR), and produce end-of-day revenue reconciliation sheets.

7.1 Processing Fee Settlements & Issuing Official Receipts

Purpose: Record payments for franchise permits, renewal fees, and violation citations.

1. From the Cashier Dashboard, locate the applicant in the **"Payment Queue"** or search by Plate Number, Body Number, or Applicant Name.
2. Verify the assessed fees: Standard MTOP regulatory fee, sticker fee, inspection fee, or penalty charges.
3. Select the payment channel: **Cash (Over-the-Counter)** or **GCash / Digital Payment**.
4. Input the **Official Receipt (OR) Number** issued by the Municipal Treasury.
5. Click **"Confirm & Release Receipt"**. The transaction is instantly recorded in the immutable treasury ledger.

✓ Expected Result: Permit status automatically advances to *"For Signing / Ready for Pickup"*, and an SMS notification is sent to the operator.

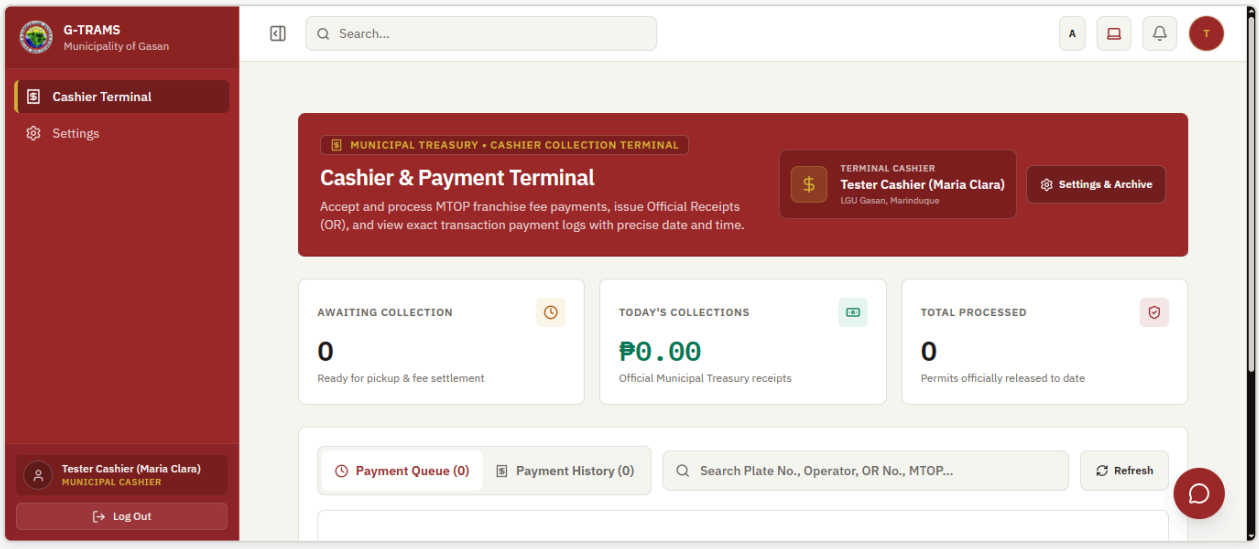


Figure 9. Municipal Cashier Payment Terminal with Real-Time Collection Counter

7.2 Generating Daily Treasury Collection & Remittance Sheet

Purpose: Produce COA-compliant daily financial turnover sheets with 1-click CSV export.

1. Click the **"Daily Remittance Sheet"** button located on the top action bar of the Cashier Terminal.
2. Review the daily collection breakdown: Total transactions, cash receipts, digital collections, and total municipal turnover.
3. Click **"Export Remittance CSV"** to download a formatted spreadsheet (`GTRAMS_Treasury_Daily_Remittance_<DATE>.csv`) for municipal accounting archives.
4. Click **"Print Remittance Form"** to generate the physical audit sheet complete with authorized signature blocks for the Cashier and Municipal Treasurer.

✓ Expected Result: End-of-day treasury reconciliation is completed in under 1 minute without manual recalculations.

8. BPLO LICENSING OFFICER PORTAL

BPLO personnel evaluate submitted documents, verify technical specifications, schedule physical inspections, and authorize final MTOP certificate issuance.

8.1 Evaluating Applications in the Review Queue

Purpose: Inspect uploaded credentials side-by-side with automated AI OCR extraction data.

1. Access the **"Franchise Approval"** module from the BPLO sidebar.
2. Click **"Review"** on any pending application.
3. Inspect the uploaded OR/CR, Cedula, Driver's License, and TODA Clearance images alongside the automated OCR field comparison.
4. If details match: Assign the scheduled physical inspection date and click **"Approve for Payment"**.
5. If documents are blurred or mismatched: Click **"Request Correction"** and type a specific note guiding the applicant.

✓ Expected Result: Applicant is notified immediately via SMS and dashboard banner regarding the review outcome.

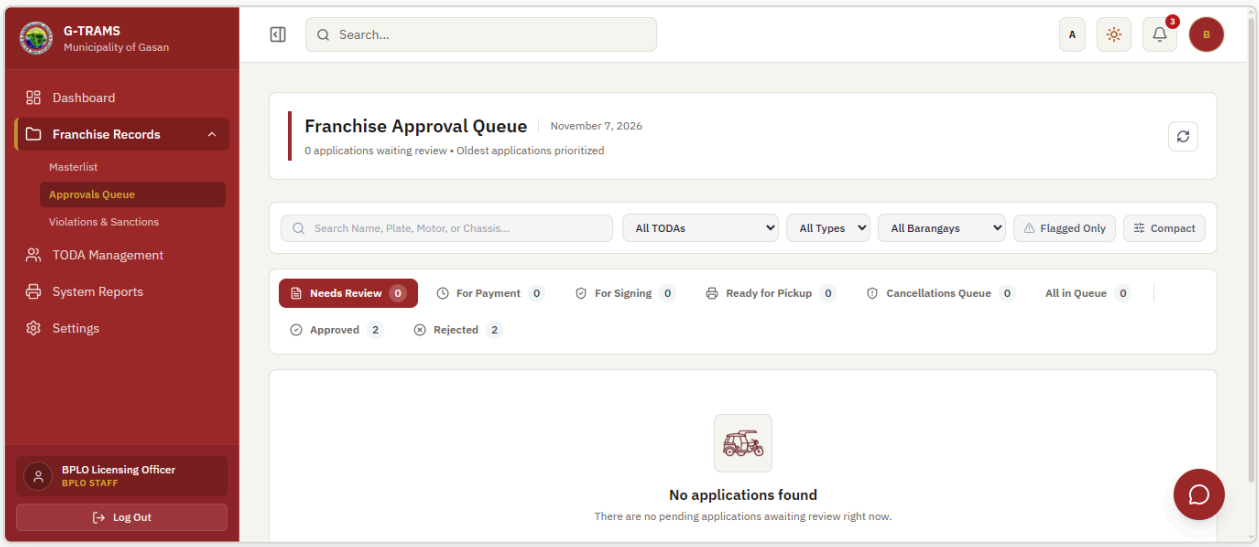


Figure 10. BPLO Application Review Queue & Document Inspection Interface

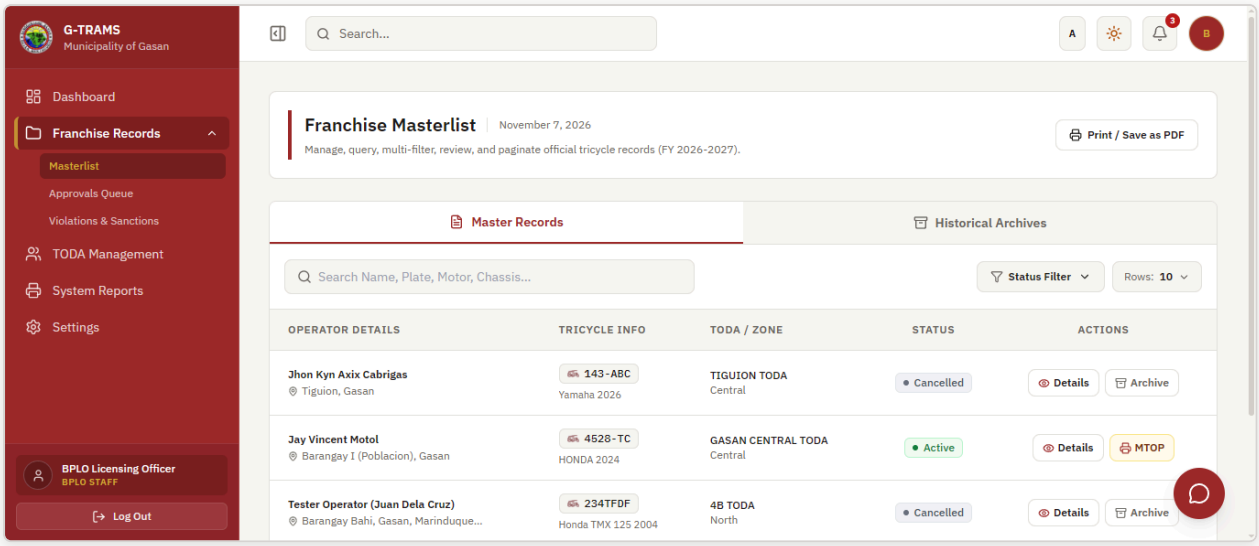


Figure 11. Municipal Franchise Masterlist Registry with Filter & Export Controls

9. SYSTEM ADMINISTRATOR PORTAL & DISASTER RECOVERY

The System Admin module provides Municipal IT staff with comprehensive user management, security audit logging, and automated database disaster recovery protocols.

9.1 Database Backup & Disaster Recovery Management (DRP)

Purpose: Safeguard municipal franchise records through automated weekly archives and instant offline downloads.

1. Navigate to **Admin Settings** and select **Tab 5: "Database Backup & Disaster Recovery"**.
2. **Automated Backup:** The system automatically performs a full compressed database backup every Sunday at 12:00 MN (Midnight) with rolling 8-week retention.
3. **Direct Offline Download:** Click **"Direct Offline Download"** to stream a fresh compressed gzip snapshot (`.json.gz`) directly to your local laptop or secure USB flash drive.
4. **Restoring Data:** In the event of server failure, select **"Upload & Restore Backup"**, choose your archive file, type `RESTORE` in the confirmation prompt, and click confirm.

✓ Expected Result: All 13 municipal database collections are restored with 100% data integrity within seconds.

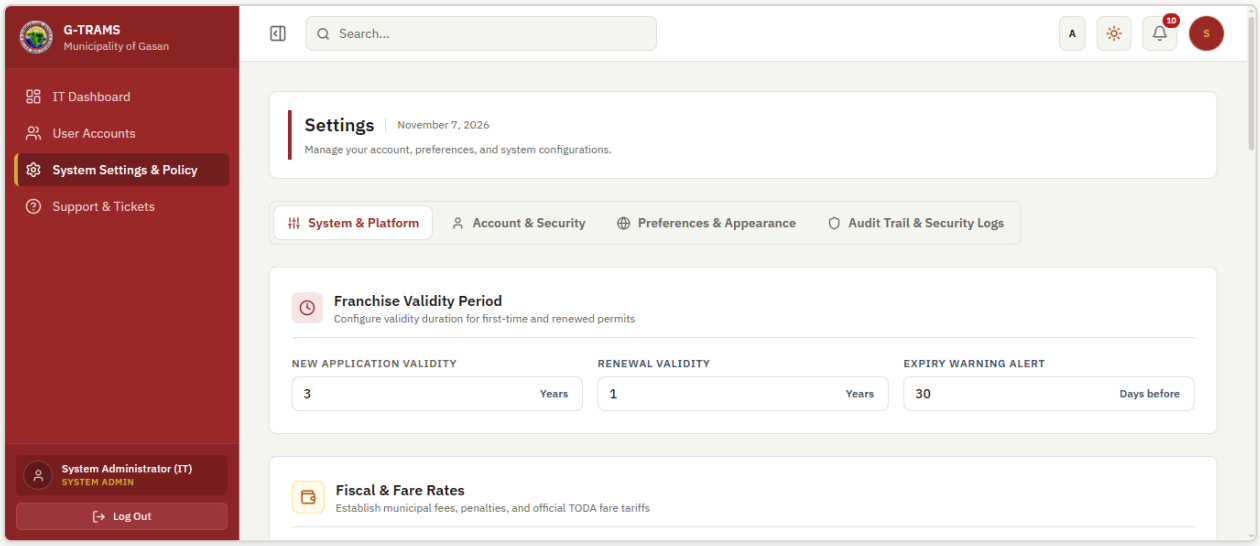


Figure 12. Database Backup & Disaster Recovery Protocol in System Admin Settings

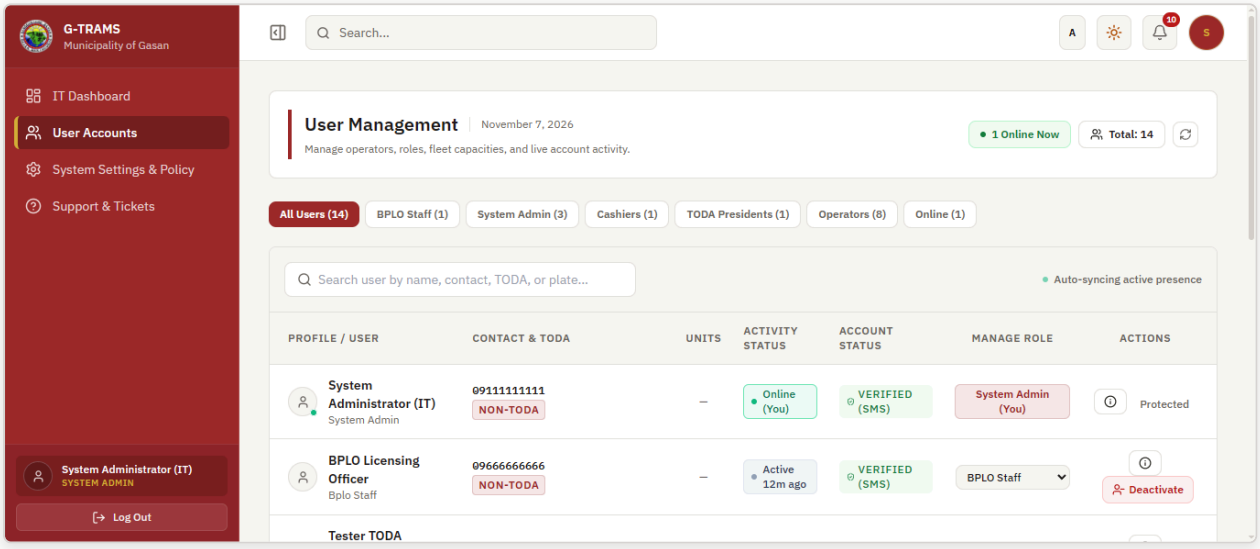


Figure 13. System User Management & Role-Based Access Control (RBAC) Interface

10. ERROR HANDLING AND TROUBLESHOOTING

The following matrix outlines common operational issues, root causes, and verified resolution steps:

Symptom / Error Message	Likely Root Cause	Step-by-Step Resolution
"Invalid credentials or incorrect password"	Typo in password or non-activated account.	Verify Caps Lock is disabled. If problem persists, click "Forgot Password?" to reset your password via SMS OTP.
"OTP code has expired or is invalid"	More than 5 minutes elapsed since code dispatch.	Click "Resend OTP". Ensure your phone has active mobile network signal in Gasan.
"Document upload failed / File too large"	File format unsupported or exceeds 10 MB limit.	Upload files in JPG, PNG, or PDF format. Reduce camera photo resolution if image exceeds 10 MB.
"Duplicate body number detected"	Another operator is already assigned this body number.	Consult your TODA President or BPLO officer to assign an unallocated municipal body number.
"Access Denied (403 Unauthorized)"	Logged-in account does not possess role permissions.	Log out and re-login with the correct authorized credentials for the specific portal module.
"Server connection timeout / Offline"	Temporary network drop or maintenance mode.	Check internet connectivity. If scheduled maintenance is underway, wait until service resumes.

11. FREQUENTLY ASKED QUESTIONS (FAQS)

Q1: How long does it take for a new franchise application to be approved?

Applications typically take **2 to 3 business days** upon submission of complete documents. You can track real-time progress on your Operator Dashboard.

Q2: Can I renew my franchise before it officially expires?

Yes. The system activates the **"Renew Franchise"** button on your dashboard **60 days prior** to your permit expiration date.

Q3: What if I lose my primary mobile phone or email address?

During registration, G-TRAMS captures a **Recovery Contact** (secondary phone or email). You can receive password reset OTPs through your registered recovery channel.

Q4: How do traffic enforcers know my digital permit is legitimate?

Every digital ID and MTOP certificate features a secure, dynamic **QR Code**. When scanned by enforcers, it retrieves the live record directly from the municipal database.

Q5: Is my personal information and document data secure?

Yes. In compliance with the **Data Privacy Act of 2012 (RA 10173)**, all passwords use one-way bcrypt hashing, active OTPs are stripped from archives, and documents are stored in encrypted cloud storage.

12. SECURITY REMINDERS & OFFICIAL SUPPORT

- **Credential Protection:** Never disclose your account password or OTP to anyone, including municipal personnel or fellow drivers.
- **Session Hygiene:** Always click **"Log Out"** when accessing the portal on shared devices, public internet cafes, or mobile kiosks.
- **Document Authenticity:** Uploading forged or altered LTO credentials constitutes a criminal violation under Philippine law and leads to permanent franchise revocation.

12.1 Technical Support & LGU Contact Channels

Office / Department	Physical Office Location	Contact Number & Email
BPLO Franchising Division	2nd Floor, Gasan Municipal Hall, Poblacion, Gasan	(042) 342-1234 • bplo@gasan.ph
Municipal Treasury Office	Ground Floor, Gasan Municipal Hall, Poblacion, Gasan	(042) 342-1235 • treasury@gasan.ph
G-TRAMS IT Technical Support	College of Information and Computing Sciences, MarSU	support@gtrams.ph • cics@mar-su.edu.ph